SET

01

Pre-NEET (2025-26)

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 26/04/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Candidate (in Capitals): _____

Roll Number (in Figures): _____

(in Words): _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ **Invigilator's Signature:** _____

Facsimile Signature Stamp of

Centre Superintendent: _____

1. Two moles of Helium gas, initially at 27°C, undergo a sudden adiabatic compression. If the work done on the gas during this process is 22.5 kJ, to what fraction of its original volume was the gas compressed? (Take the universal gas constant $R = \frac{25}{3}$ J mol^{-1} K^{-1})

(1) $\frac{1}{2}$ (2) $\frac{1}{4}$

(3) $\frac{1}{8}$ (4) $\frac{1}{16}$

2. Match the physical quantities in **List-I** to their SI unit expressions provided in **List-II**. (Symbol have usual meaning)

List-I		List-II	
(A)	Magnetic Flux (ϕ)	(I)	$C \cdot V^{-1} \cdot m^{-1}$
(B)	Self-Inductance (L)	(II)	$m^2 \cdot V^{-1} \cdot s^{-1}$
(C)	Permittivity of free space (ϵ_0)	(III)	$J \cdot A^{-1}$
(D)	Mobility of charge carriers (μ)	(IV)	$\Omega \cdot s$

Choose the **correct** answer from the options given below:

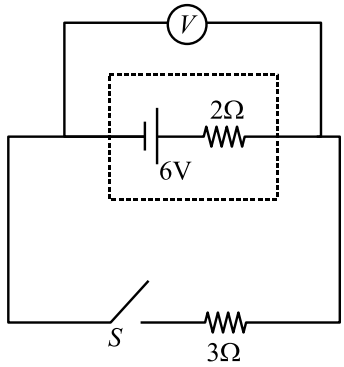
(1) A-IV, B-III, C-II, D-I

(2) A-III, B-I, C-IV, D-II

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-IV, C-I, D-III

3. In the circuit shown in the figure, the cell has an electromotive force (emf) of 6.0 V and an internal resistance of 2.0 Ω. The external resistance is 3.0 Ω. What is the reading of the ideal voltmeter when the switch S is open, and what is its reading when the switch S is closed, respectively?



(1) 0.0 V and 3.6 V

(2) 6.0 V and 2.4 V

(3) 6.0 V and 6.0 V

(4) 6.0 V and 3.6 V

4. An acoustic engineer detects that the fundamental frequency of a simple closed organ pipe is 250 Hz. What will be the frequency of its second overtone?

(1) 500 Hz (2) 750 Hz

(3) 1000 Hz (4) 1250 Hz

5. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): In a hydrogen atom, the magnitude of the potential energy of the revolving electron is exactly same as the magnitude of its kinetic energy, and its total energy is negative.

Reason (R): The electrostatic force of attraction between the nucleus and the electron provides the necessary centripetal force for the circular orbit.

In the light of the above statements, choose the **correct** answer from the options given below:

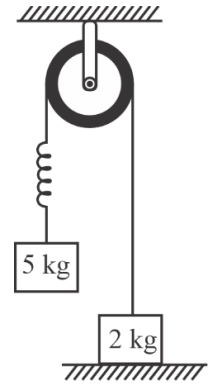
(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

6. The system shown below, having a spring which is initially unstretched and is connected to a block with a string passing through a pulley, is released from rest. Pulley and spring are massless and the friction is absent everywhere. The speed of 5 kg block, at the instant 2 kg block leaves the contact with ground is: (take force constant of the spring $K = 40$ N/m and $g = 10$ m/s^2)



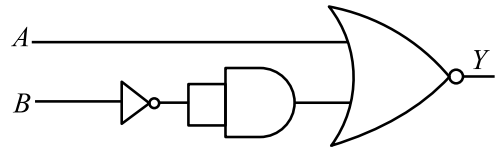
(1) $\sqrt{2}$ m/s

(2) $2\sqrt{2}$ m/s

(3) 2 m/s

(4) $4\sqrt{2}$ m/s

7. A combination of gates is shown below. Which of the options given below represent the **correct** Boolean expression of the output Y.



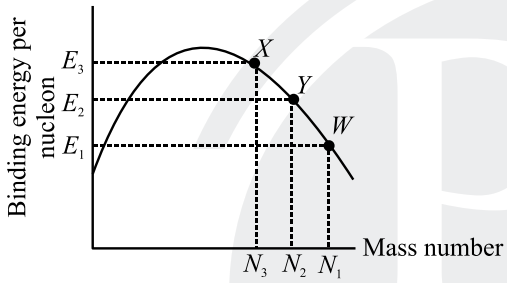
(1) $Y = A + \overline{B}$

(2) $Y = \overline{A} + B$

(3) $Y = A \cdot \overline{B}$

(4) $Y = \overline{A} \cdot B$

8. Consider the nuclear fission reaction: $W \rightarrow X + Y$. What is the Q -value (energy released) of the reaction?



(1) $E_1N_1 - (E_2N_2 + E_3N_3)$

(2) $E_2N_2 + E_3N_3 - E_1N_1$

(3) $E_2N_2 + E_1N_1 - E_3N_3$

(4) $E_1N_1 + E_3N_3 - E_2N_2$

9. A solid sphere and a thin hollow sphere of same mass and same external radius are rotating about their respective diameters. If both possess identical rotational kinetic energy, what is the ratio of the angular momentum of the solid sphere to that of the hollow sphere?

(1) $\sqrt{\frac{5}{3}}$ (2) $\frac{3}{5}$

(3) $\sqrt{\frac{3}{5}}$ (4) $\frac{5}{3}$

10. A technician tests a block of an unknown homogeneous magnetic material and records its relative permeability to be 0.9983 at room temperature. The examined material is:

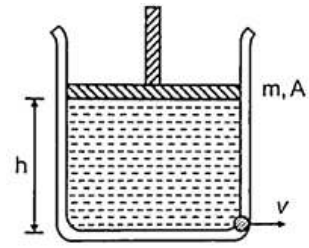
(1) Paramagnetic

(2) Diamagnetic

(3) Ferromagnetic

(4) Non-magnetic

11. A cylindrical vessel contains a liquid of density ρ upto a height h. The liquid is closed by a piston of mass m and area of cross-section A. There is a small hole at the bottom of the vessel. The speed v with which the liquid comes out of the hole is:



(1) $\sqrt{2gh}$

(2) $\sqrt{2\left(gh + \frac{mg}{\rho A}\right)}$

(3) $\sqrt{2\left(gh + \frac{mg}{A}\right)}$

(4) $\sqrt{2gh + \frac{mg}{A}}$

12. A long straight wire carries a current of 10 A. An electron is projected with a velocity of 10^5 m/s parallel to the wire at a perpendicular distance of 0.2 m from it. Calculate the magnitude of the magnetic force experienced by the electron.

(1) 1.6×10^{-18} N

(2) 3.2×10^{-19} N

(3) 1.6×10^{-19} N

(4) 8.0×10^{-20} N

13. A point particle executing Simple Harmonic Motion (SHM) along the x-axis has a complete time period of T seconds. The peak amplitude of its oscillation is A. The particle is released from the mean position ($x = 0$) at $t = 0$. It takes time t_1 to move from $x = 0$ to $x = +\frac{A}{2}$, and a subsequent time t_2 to continue from $x = +\frac{A}{2}$ up to the extreme $x = A$. Determine the ratio $t_1 : t_2$.

(1) 1: 1 (2) 2: 1

(3) 1: 2 (4) 1: 3

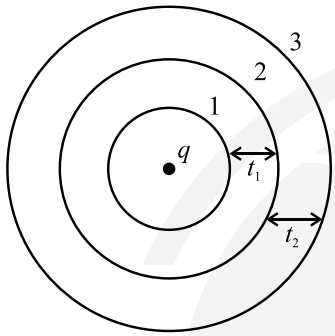
[Contd...

14. Two distinct open organ pipes measuring 50 cm and 50.5 cm in length respectively produce 3 beats per second when sounded together in their fundamental mode of vibration. Based on this observation, what is the velocity of sound in the air?

(1) 330 m/s (2) 303 m/s

(3) 300 m/s (4) 340 m/s

15. Figure shows three spherical and equipotential surfaces 1, 2 and 3 generated due to a point charge q. The potential difference $V_1 - V_2 = V_2 - V_3$. If t_1 and t_2 be the distance between them, then:



(1) $t_1 = t_2$ (2) $t_1 > t_2$

(3) $t_1 < t_2$ (4) $t_1 \le t_2$

16. An isolated parallel plate capacitor with plain air between its metallic plates boasts a total capacitance C. If the intervening space between the plates is half-filled with a uniform dielectric slab of constant $K = 3$ (introduced parallel to the plates, filling exactly half the vertical thickness), what will be the new effective capacitance?

(1) $1.5C$ (2) $2C$

(3) $3C$ (4) $4C$

17. Consider the following two statements concerning thermodynamics.

Statement I: The internal energy of an ideal gas depends solely on its absolute temperature.

Statement II: During an isothermal expansion of an ideal gas, the heat supplied to the gas is completely converted into macroscopic mechanical work.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

18. Four hypothetical planets $P_1, P_2, P_3,$ and P_4 are composed of different uniform materials and have the same total mass, but their radii are in the ratio 1 : 2 : 3 : 4. What is the ratio of the acceleration due to gravity (g_d) at a depth equal to half of their respective radii

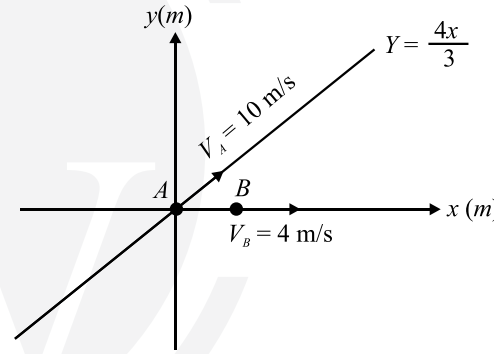
(1) 1 : 2 : 3 : 4

(2) 16 : 9 : 4 : 1

(3) 144 : 36 : 16 : 9

(4) 1 : 4 : 9 : 16

19. A particle A of mass 2 kg is moving along the line $y = \frac{4x}{3}$ as shown in the figure with speed 10 m/s. Another particle B of mass 2 kg moves along + ve x-axis with speed 4 m/s. The speed of C.O.M. of the system of two particles A and B is:



(1) $\frac{13}{2}$ m/s

(2) $\sqrt{41}$ m/s

(3) $\sqrt{45}$ m/s

(4) 5 m/s

20. In a classic Young's double-slit experiment (YDSE) setup, if the primary monochromatic light source is slightly shifted physically upwards by a small distance y in a plane parallel to the plane containing the slits, what behavior will the central maximum of the interference pattern exhibit on the screen?

(1) Shift upwards

(2) Shift downwards

(3) Remain completely unchanged

(4) Disappear completely

21. A heavy, uniform rigid rod of length L and mass M is securely hinged via a frictionless pivot at one of its ends and is temporarily held perfectly horizontal. The rod is then abruptly released from this horizontal position. Determine the instantaneous angular acceleration of the rod at the moment it is released. (g is acceleration due to gravity)

(1) $\frac{g}{L}$

(2) $\frac{2g}{L}$

(3) $\frac{3g}{2L}$

(4) $\frac{3g}{L}$

22. Consider an unbiased p-n junction diode at room temperature. Its depletion region has a width of $1\,\mu m$ and a built-in potential barrier of 0.7 V. An electron enters this depletion region from the p-side with an initial kinetic energy of 0.5 eV. Assuming the electron does not suffer any internal collisions along its path, what will be its kinetic energy when it reaches the n-side?

(1) 1.2 eV (2) 0.5 eV

(3) 0.2 eV (4) Zero

23. A galvanometer having a resistance of $50\,\Omega$ gives a full scale deflection for a current of 0.05 A. The approximate length (in metres) of a resistance wire of area of cross section $3 \times 10^{-2}\,cm^2$ that can be used to convert the galvanometer into an ammeter which can read a maximum of 5 A current is (Specific resistance of the wire $\rho = 5 \times 10^{-7}\ \Omega\,m$)

(1) 9 (2) 6

(3) 3 (4) 1.5

24. An electric motor is used to lift an elevator upwards at a constant velocity of 2 m/s. The total mass of the elevator, including its passengers, is 1000 kg. If a constant kinetic frictional force of 4000 N opposes its upward motion, calculate the total mechanical power delivered by the electric motor. (Take $g = 10\ m/s^2$)

(1) 14 kW (2) 20 kW

(3) 24 kW (4) 28 kW

25. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The angle of repose for a block on an inclined plane is numerically equal to the angle of friction.

Reason (R): The maximum static friction force between two rough surfaces is entirely independent of the normal reaction acting between them.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

26. Two distinct, stable atomic nuclei have their respective mass numbers in the ratio of 1: 8. What is the resulting ratio of their respective nuclear densities?

(1) 1: 2

(2) 1: 8

(3) 1: 1

(4) 8: 1

27. Two monochromatic light beams of frequencies v and $2v$ are incident on a given photosensitive metal surface independently. If the ratio of the maximum kinetic energies of the emitted photoelectrons in the two cases is observed to be 1: 3, what is the threshold frequency of the metal?

(1) $\frac{v}{3}$

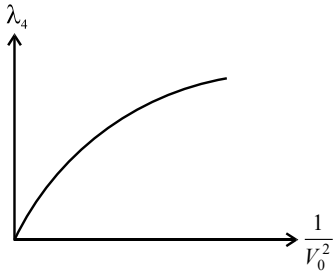
(2) $\frac{v}{2}$

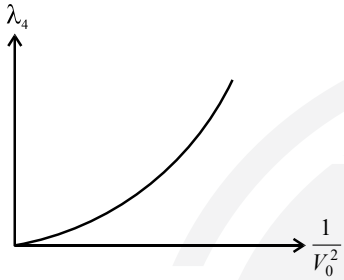
(3) $\frac{2v}{3}$

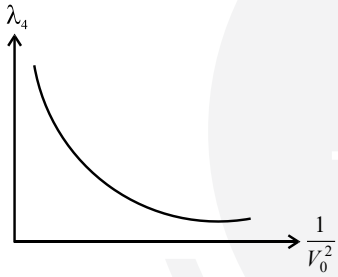
(4) $\frac{3v}{4}$

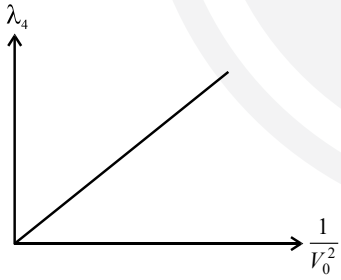
[Contd...

28. In a photoelectric effect experiment, monochromatic light is incident on a metal surface. If V_0 is the stopping potential and λ is the de-Broglie wavelength of the emitted photoelectrons possessing maximum kinetic energy, which of the following graphs correctly represents the variation of λ^4 with $\frac{1}{V_0^2}$?

(1) 

(2) 

(3) 

(4) 

29. Two coaxial coils have a mutual inductance of 1.5×10^{-3} H. If an average emf of 0.6 V is induced in the secondary coil over a time interval of 0.02 s, what is the magnitude of the total change in current in the primary coil?

(1) 4 A (2) 8 A
(3) 2 A (4) 16 A

30. An artificial satellite is placed into a stable circular orbit around the Earth. The orbital speed of this satellite is found to be equal to half the magnitude of the escape velocity from the Earth's surface. What is the height of this satellite above the Earth's surface? (R is the radius of the Earth).

(1) $\frac{R}{2}$ (2) $2R$
(3) $4R$ (4) R

31. At what absolute temperature scale reading is the precise root mean square (rms) speed of molecular hydrogen (H_2) matches with the standard rms speed of molecular oxygen (O_2) maintained at 320 K?

(1) 20 K (2) 40 K
(3) 10 K (4) 80 K

32. Four different AC transformers are tested under load. Based on the measured values of primary voltage (V_p), primary current (I_p), secondary voltage (V_s), and secondary current (I_s), which transformer has the minimum efficiency?

(1) $V_p = 200$ V, $I_p = 5$ A, $V_s = 100$ V, $I_s = 9$ A
(2) $V_p = 220$ V, $I_p = 10$ A, $V_s = 22$ V, $I_s = 85$ A
(3) $V_p = 100$ V, $I_p = 2$ A, $V_s = 50$ V, $I_s = 3$ A
(4) $V_p = 500$ V, $I_p = 4$ A, $V_s = 100$ V, $I_s = 19$ A

33. In a hypothetical new system of physical units, if linear velocity v, linear acceleration A, and absolute mechanical force F are chosen as the primary fundamental quantities, what would be the derived dimensional formula for mass?

(1) $[FA^{-1}V^0]$ (2) $[FVA^{-1}]$
(3) $[FAV^{-2}]$ (4) $[F^2A^{-1}V^0]$

34. A monochromatic plane electromagnetic wave travels from vacuum into a non-magnetic medium ($\mu_r = 1$) that has a refractive index of $n = 4$. Assuming that the average intensity of the wave remains the same as it enters the medium, what is the ratio of the Electric Field amplitude in vacuum (E_{vac}) to the Electric Field amplitude in the medium (E_{med})?

(1) 4 : 1 (2) 1 : 4
(3) 2 : 1 (4) 1 : 1

35. An object is placed at a distance of 30.0 cm from a spherical mirror on the principal axis. The absolute value of lateral magnification is $\frac{1}{3}$. If the image produced is inverted, then the focal length of the mirror is:

(1) 15.0 cm (2) 7.5 cm
(3) –15.0 cm (4) –7.5 cm

[Contd...

36. A magnetizing field H of 500 A/m produces a magnetic flux of 4×10^{-5} Wb in an iron rod. If the area of cross-section of the rod is 0.5 cm^2, then permeability of the rod will be:

(1) 1×10^{-3} H/m (2) 2.3×10^{-4} H/m

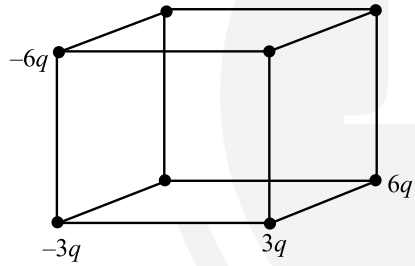
(3) 3×10^{-5} H/m (4) 1.6×10^{-3} H/m

37. In a series LCR circuit powered by an AC source, precise voltmeters read the potential differences across the pure inductor, the capacitor, and the non-inductive resistor as 20 V, 20 V, and 40 V respectively. What is the fundamental phase difference between the total applied source voltage and the main current flowing in the circuit?

(1) 30° (2) 45°

(3) 90° (4) 0°

38. Different point charges are placed at the vertices of a cube as shown.

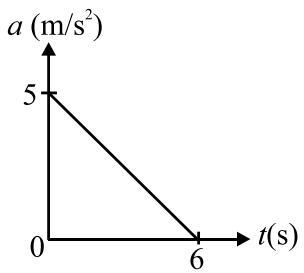


Net electric flux through the cube is:

(1) $\frac{3q}{\epsilon_0}$ (2) $\frac{-3q}{\epsilon_0}$

(3) $\frac{4q}{\epsilon_0}$ (4) Zero

39. A particle starts from rest. Its acceleration at time $t=0$ is 5 m/s^2 which varies with time as shown in the figure. The maximum speed of the particle will be:



(1) 7.5 m/s (2) 15 m/s

(3) 30 m/s (4) 37.5 m/s

40. A projectile is fired from the origin over a horizontal surface at an angle θ with the horizontal. For the velocity vector to be strictly perpendicular to the position vector of the projectile at some instant of its flight, the angle of projection θ must satisfy a minimum condition. What is this condition?

(1) $\sin\theta\geq\sqrt{\frac{2}{3}}$ (2) $\sin\theta\geq\frac{2\sqrt{2}}{3}$

(3) $\cos\theta\geq\sqrt{\frac{1}{3}}$ (4) $\tan\theta\geq\sqrt{8}$

41. A uniform aluminium sphere of 20 cm diameter is heated from 50°C to 150°C. Its volume changes by (Coefficient of linear expansion for aluminium $(\alpha_{Al}=23\times10^{-6}\ ^\circ C^{-1})$)

(1) 28.9 cc (2) 2.89 cc

(3) 0.289 cc (4) 9.28 cc

42. A 3 cm × 10 cm rectangular coil with number of turns 300 is placed in uniform magnetic field of strength 0.2 T. The maximum magnetic torque acting on it when a current of 10^{-4} A is passed through it:

(1) 3.6×10^{-6} Nm (2) 2.4×10^{-5} Nm

(3) 2.4×10^{-6} Nm (4) 1.8×10^{-5} Nm

43. A uniform electrical wire of total resistance $40\,\Omega$ is carefully bent to form a perfect square. A DC battery of EMF 20 V and internal resistance $2.5\,\Omega$ is connected across the two ends of exactly one side of the square. Calculate the electrical power dissipated in the other three sides of the square combined.

(1) 5 W (2) 7.5 W

(3) 10 W (4) 15 W

44. The equation of a standing wave produced on a string fixed at both ends is

$y=0.4\sin[(0.314\ \text{cm}^{-1})x]\times\cos[(600\pi\ \text{s}^{-1})t]$

(where x and y are in cm and t is in s)

The smallest possible length of the string for this will be:

(1) 20 cm (2) 40 cm

(3) 10 cm (4) 5 cm

45. A stationary explosive device on a frictionless surface detonates into three fragments. Two fragments, each of mass 3 kg, fly off at right angles to each other, both with a speed of 4 m/s. If the third fragment moves with a speed of $6\sqrt{2}$ m/s, calculate the total original mass of the device before the blast.

(1) 6 kg (2) 8 kg

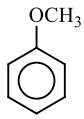
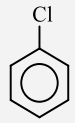
(3) 10 kg (4) 12 kg

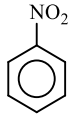
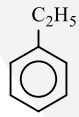
[Contd...

46. Identify the **correct** statements:
A. Steroid hormones are produced by adrenal cortex and gonads
B. Insulin is released in response to the rapid lowering in blood glucose level
C. Epinephrine and norepinephrine mediate responses to external stimuli.
D. Abnormally high level of thyroxine leads to hypothyroidism which is characterised by lethargyness and obesity.
E. glucocorticoids control the carbohydrate metabolism, modulate inflammatory reactions
(1) A, B and C (2) A, C and E
(3) B, D and C (4) A, B and E

47. 2 g of a nitrogen containing compound give 0.17 g of NH_3 in Kjeldahl method. The percentage of nitrogen in organic compound is:
(1) 9% (2) 7%
(3) 13% (4) 17%

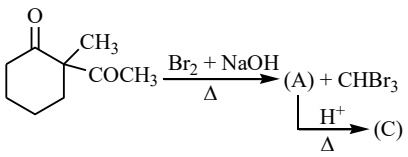
48. For the following compounds, the decreasing order of reactivity towards electrophilic substitution reaction is:

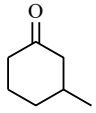
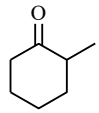
(I)  (II) 

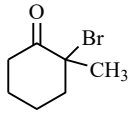
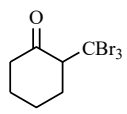
(III)  (IV) 

(1) III > I > IV > II (2) I > III > IV > II
(3) I > IV > II > III (4) IV > I > II > III

49. Among $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, O_2, $Na_3[CoF_6]$, Na_2O_2 and CsO_2, the total number of paramagnetic compounds is:
(1) 2 (2) 3
(3) 4 (4) 5

50.  : Product (C) of the reaction is:

(1)  (2) 

(3)  (4) 

51. On passing two faraday of electricity through dil. HCl, the volume of gas evolve at anode at STP is:
(1) O_2, 22.4 L (2) Cl_2, 22.4 L
(3) H_2, 22.4 L (4) Cl_2, 11.2 L

52. Identify the **incorrect** statements:
$[-\log(4.525 \times 10^{-3}) = 2.34, -\log(0.0162) = 1.79, -\log(0.0375) = 1.43, -\log(0.0136) = 1.87]$
A. pOH of 2 g of TlOH dissolved in water to give 2 litre of solution is 2.34
B. pH of 0.3 g of $Ca(OH)_2$ dissolved in water to give 500 mL of solution is 1.79.
C. pOH of 0.3 g of NaOH dissolved in water to give 200 mL of solution is 1.43.
D. pOH of 1mL of 13.6 M HCl is diluted with water to give 1 litre of solution is 1.87.
(1) A and C (2) A and D
(3) B and C (4) B and D

53. Match **List-I** with **List-II**.

List-I (Molecule)		List-II (Lone pair of central atom)	
(A)	BrF_5	(I)	3
(B)	PF_5	(II)	2
(C)	SF_2	(III)	1
(D)	XeF_2	(IV)	0

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-II, B-IV, C-III, D-I

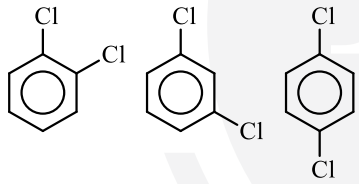
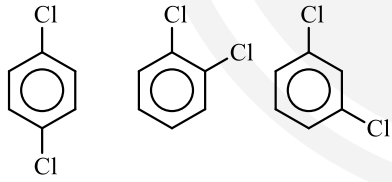
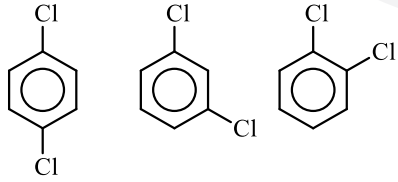
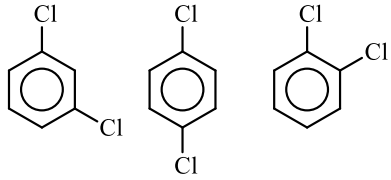
54. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;
Assertion A: Vapour pressure of 0.1 M urea solution is more than that of 0.2 M urea solution.
Reason R: Vapour pressure is a colligative property.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true but R is NOT the correct explanation of A.

55. If bond energy of C–H, H–H and C=C bonds are 414, 436 and 611 kJ mol^{-1} respectively then $\Delta_f H°$ of $CH_2=CH_2$ will be:
(Given: $\Delta_{sub}H°$ of C(graphite) is 710 kJ mol^{-1})
(1) 30 kJ mol^{-1} (2) 20 kJ mol^{-1}
(3) 25 kJ mol^{-1} (4) 35 kJ mol^{-1}

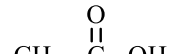
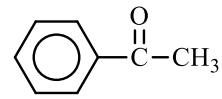
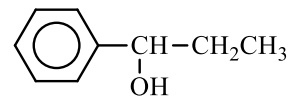
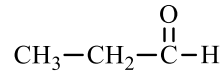
[Contd...

56. The pH of a buffer solution containing HA and NaA is 5. If the K_a of HA is 10^{-6}, the ratio of salt to acid concentration in the buffer solution will be:
(1) 0.1
(2) 1.1
(3) 10
(4) 1

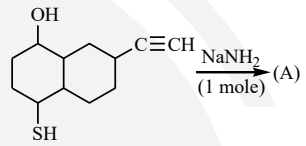
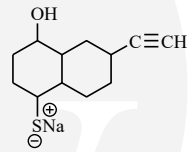
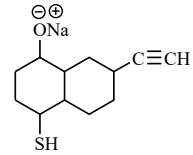
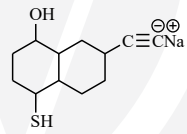
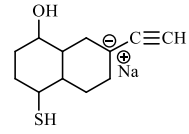
57. Given below are two statements:
Statement I: On addition of starch after the liberation of iodine from the reaction of Cu^{2+} ions on iodide ions, an intense blue colour appears.
Statement II: $KMnO_4$ acts as the self indicator.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

58. The **correct** order of melting point of dichlorobenzenes is:
(1) 
(2) 
(3) 
(4) 

59. The major product obtained on reaction of phenol with sodium hydroxide and carbon dioxide is:
(1) Salicyaldehyde
(2) Terephthalic acid
(3) Salicylic acid
(4) Cresol

60. Which among the following compounds give yellow solid when reacted with NaOI/NaOH?
(1) 
(2) 
(3) 
(4) 

61. The reagent that can selectively precipitate S^{2-} from a mixture of S^{2-} and SO_4^{2-} in aqueous solution is:
(1) $BaCl_2$ (2) $CuCl_2$
(3) $(CH_3COO)_2Pb$ (4) $Na_2[Fe(CN)_5NO]$

62.  ; Product (A) is:
(1) 
(2) 
(3) 
(4) 

63. Consider the following first order competing reactions:
$X \xrightarrow{k_1} A + B$ and $Y \xrightarrow{k_2} C + D$
If 50% of the reaction of X was completed when 96% of the reaction of Y was completed, the ratio of their rate constants (k_2/k_1) is:
(1) 4.06 (2) 0.215
(3) 1.1 (4) 4.65

64. Given the listed standard electrode potentials, what is E° for the cell:
$4BiO^+(aq) + 3N_2H_5^+(aq) \rightarrow 4Bi(s) + 3N_2(g) + 4H_2O(\ell) + 7H^+(aq)$
$N_2(g) + 5H^+(aq) + 4e^- \rightarrow N_2H_5^+(aq), E^o = -0.23\ V$
$BiO^+(aq) + 2H^+(aq) + 3e^- \rightarrow Bi(s) + H_2O(\ell), E^o = +0.32\ V$
(1) +0.55 V (2) +0.34 V
(3) +1.88 V (4) +0.09 V

[Contd...

65. The **correct** statements among the following are:
(A) Tl^{3+} is a powerful oxidising agent than Al^{3+}
(B) The metallic radius of Al is less than Ga.
(C) Density of B is more than Ga.
(D) Electronegativity of Tl is more than Al
(1) B and C (2) A and C
(3) A and D (4) B and D

66. Given below are two statements:
Statement I: Tertiary amines react with benzenesulphonyl chloride very easily.
Statement II: In Friedel-Crafts alkylation of aniline, the alkylated product at ortho and para position are obtain.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

67. Identify the essential amino acids from below:
(A) Leucine (B) Proline
(C) Tryptophan (D) Arginine
(E) Histidine
Choose the **correct** answer from the options given below:
(1) A, B and C only (2) B, C and D only
(3) C, D and E only (4) A, C and E only

68. Match **List-I** with **List-II**.

List-I (Element)		List-II (Electro gain enthalpy in kJ/mol)	
(A)	O	(I)	–141
(B)	S	(II)	–200
(C)	Cl	(III)	–349
(D)	Te	(IV)	–190

Choose the **correct** answer from the options given below:
(1) A-IV, B-I, C-II, D-III
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-I, B-II, C-III, D-IV

69. Given below are two statements:
Statement I: The de Broglie wavelength of a hockey ball of mass 0.1 kg is 1.52×10^{-39} m it moves with the same speed as a proton whose speed after being accelerated through 1000 V is 4.37×10^{5} m s^{-1}.
Statement II: According to de Broglie, every object in motion has a wave character
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

70. A certain public water supply contains 0.10 ppb (part per billion) of chloroform ($CHCl_3$). How many molecules of $CHCl_3$ would be obtained in 0.478 mL drop of this water? (assumed d = 1 g/mL)
(1) $4 \times 10^{-13} \times N_A$ (2) $10^{-3} \times N_A$
(3) $4 \times 10^{-10} \times N_A$ (4) $4 \times 10^{-5} \times N_A$

71. For the reaction $2HI(g) \rightarrow H_2(g) + I_2(g)$
the activation energy is 209.5 kJ/mol at 581 K.
Given below are two statements:
Statement I: The fraction of reactant molecules having energy greater than or equal to the activation energy is numerically very small at 581 K.
Statement II: This is because the value of $\frac{E_a}{RT}$ is large, so the Boltzmann factor $e^{-E_a/RT}$ becomes very small.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement-I and Statement-II are correct.
(2) Both Statement-I and Statement-II are incorrect.
(3) Statement-I is correct but Statement-II is incorrect.
(4) Statement-II is correct but Statement-I is incorrect.

72. The **correct** statement(s) regarding the binary transition metal carbonyl compounds is (are) (Atomic numbers: Fe = 26, Ni = 28)
A. Total number of valence shell electrons at metal centre in $Fe(CO)_5$ or $Ni(CO)_4$ is 16
B. These are predominantly low spin in nature
C. Metal-carbon bond strengthens when the oxidation state of the metal is lowered
D The carbonyl C–O bond weakens when the oxidation state of the metal is increased
(1) A, B and C only
(2) B, C only
(3) A, C and D only
(4) B, C and D only

[Contd...

73. Assume that the decomposition of HNO_3 can be represented by the following equation

$$4HNO_3(g) \rightleftharpoons 4NO_2(g) + 2H_2O(g) + O_2(g)$$

and the reaction approaches equilibrium at 400 K temperature and 30 atm pressure. At equilibrium partial pressure of HNO_3 is 2 atm. Calculate K_c in $(mol/L)^3$ at 400 K: (Use: R = 0.08 atm-L/mol-K)

(1) 4 (2) 8
(3) 16 (4) 32

74. When SO_2 is passed into an acidified potassium dichromate solution, the oxidation numbers of sulphur and chromium in the final products respectively are:

(1) +6, +6 (2) +6, +3
(3) 0, +3 (4) +2, +3

75. Which of the following best describes the diagram of given molecular orbital?

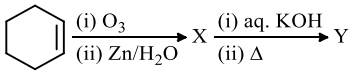


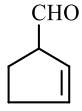
(1) A non-bonding orbital
(2) An antibonding σ-orbital
(3) A bonding π-orbital
(4) An antibonding π-orbital

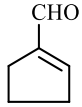
76. When 10 g of $CH_3CH_2CHClCOOH$ is dissolved in 250 g of water, the solution behaves as a weak electrolyte.
(Given: $K_a = 1.4 \times 10^{-3}$, $K_f = 1.86$ K kg mol^{-1})
The van't Hoff factor i for the solution is:

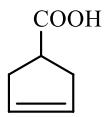
(1) 0.94 (2) 1.07
(3) 1.40 (4) 1.86

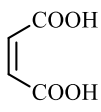
77. The major product (Y) in the following reaction sequence is:



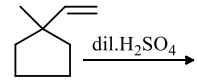
(1) 

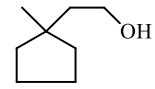
(2) 

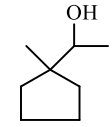
(3) 

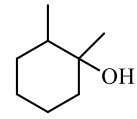
(4) 

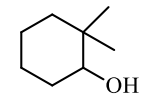
78. The major product of the following reaction is:



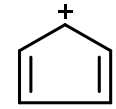
(1) 

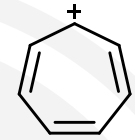
(2) 

(3) 

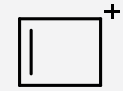
(4) 

79. Most stable carbocation among the following is:

(1) 

(2) 

(3) 

(4) 

80. Given below are two statements:
Statement I: The tanks used by scuba divers are filled with air diluted with helium.
Statement II: An azeotropic mixture of ethanol and water contains approximately 95% by volume of ethanol.
In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

81. Which of the following species have a different molecular shape from NH_3?

(I) SeO_3^{2-}
(II) CH_3^-
(III) BH_3
(IV) CH_3^+

(1) I and II only (2) III and IV only
(3) I and III only (4) II and IV only

[Contd...

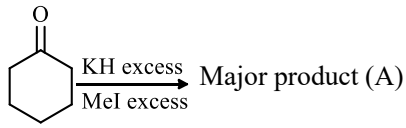
82. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R;
Assertion A: When 10 g of element 'A' (molar mass = 30 g mol^{-1}) gains electrons to form A^- ions, 23 kcal of energy is released.
Reason R: Electron gain enthalpy expressed in –3 eV per atom can be converted to kcal mol^{-1} using the relation: 1 eV per atom = 23 kcal mol^{-1}.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is NOT the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

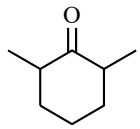
83. The highest boiling point among the following is of:
(1) He (2) Ar
(3) Kr (4) Xe

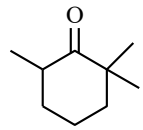
84. Match **List-I** with **List-II**.

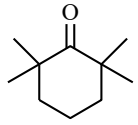
List-I (Precipitating reagent)		List-II (Metal ion)	
(A)	Passing H_2S in presence of dil. HCl	(I)	Cu^{2+}
(B)	NH_4OH in presence of NH_4Cl	(II)	Al^{3+}
(C)	$(NH_4)_2CO_3$ in presence of NH_4OH	(III)	Mn^{2+}
(D)	Passing H_2S in presence of NH_4OH	(IV)	Ba^{2+}

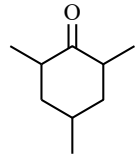
Choose the **correct** answer from the options given below:
(1) A-I, B-II, C-IV, D-III
(2) A-I, B-III, C-IV, D-II
(3) A-III, B-IV, C-II, D-I
(4) A-IV, B-III, C-II, D-I

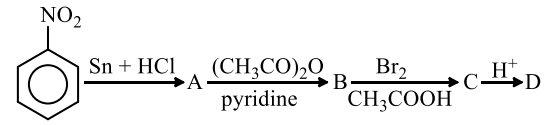
85. 

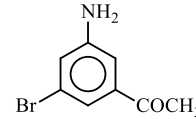
Major product (A)
Product (A) is:
(1) 

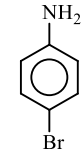
(2) 

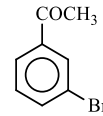
(3) 

(4) 

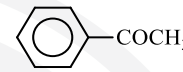
86. The major product (D) of the following reaction sequence is:


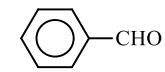
(1) 

(2) 

(3) 

(4) 

87. Which of the following compounds gives both a positive Tollen's test and a positive Fehling's test?
(1) 

(2) 

(3) HCHO (4) CH_3–COOH

88. Given below are two statements:
Statement I: The highest fluoride of Mn is MnF_7.
Statement II: The actinoid contraction is due to poor shielding by 6f electrons.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

89. The **correct** order of electronegativity for the given elements on the Pauling scale is:
(1) F > N > O (2) Cl > S > P
(3) Mg > Be > Na (4) O > C > Cl

90. An organic compound contains 69.77% carbon, 11.63% hydrogen and rest oxygen. The molecular mass of the compound is 86. It does not reduce Tollens' reagent but forms an addition compound with sodium hydrogensulphite and does not give positive iodoform test. On vigorous oxidation it gives ethanoic and propanoic acid. Name the possible structure of the compound.
(1) pentanal
(2) pentan-2-one
(3) pentan-3-one
(4) butanone

[Contd...

91. If proteases and nucleases were present freely in the blood, what would **most** likely happen?
(1) Blood clotting would become faster due to activation of platelets
(2) Proteins and nucleic acids in blood would be continuously degraded, leading to severe damage
(3) Only glucose metabolism would be affected, while proteins remain stable
(4) There would be no effect because blood contains natural inhibitors for all enzymes

92. Given below are two statements:
Statement I: In pteridophytes, the main plant body is a sporophyte called prothallus.
Statement II: The sex organs in bryophytes develop in the sporophytic phase of life cycle.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

93. A scientist is growing C_3 plants in a controlled environment. If she doubles the CO_2 concentration in the chamber while keeping light intensity and temperature optimal, what is the most likely outcome on the rate of photosynthesis?
(1) The rate will decrease because high CO_2 is toxic.
(2) The rate will show a significant increase as CO_2 is often a limiting factor for C_3 plants.
(3) The rate will remain unchanged because light is the primary limiting factor.
(4) The rate will increase initially and then decrease sharply due to photorespiration.

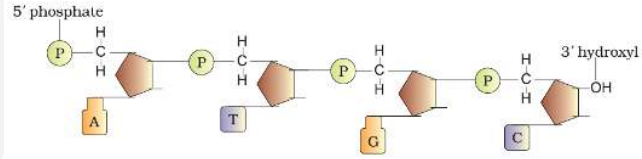
94. Which of the following cannot take place in the cell cycle of a male honey bee?
A. Replication of DNA.
B. Production of haploid cells.
C. Recombination between homologous chromosomes.
D. Duplication of centriole.
E. Formation of a cell plate.
Choose the **correct** option.
(1) A and D (2) B and C
(3) A and C (4) C and E

95. At what specific stage of its complex life cycle does the *Plasmodium* physically enter the human body?
(1) When toxic haemozoin is released into the blood.
(2) When it multiplies actively within the liver cells.
(3) When sporozoites are injected into the bloodstream through the bite of an infected female *Anopheles* mosquito.
(4) When sexually mature gametocytes develop heavily inside red blood cells.

96. A singular, un-paired, U-shaped bone present at the base of the buccal cavity is;
(1) Mandible
(2) Occipital condyle
(3) Sphenoid
(4) Hyoid

97. The value of regression coefficient for fruit-eating birds in tropical forests of different continents is;
(1) 1.15 (2) 0.12
(3) 1.2 (4) 0.15

98. Given below is a fragment of polynucleotide chain.



If it forms a double helix, then the total number of hydrogen bonds formed between the two strands in this fragment will be;
(1) 10
(2) 8
(3) 12
(4) 6

99. Identify the **correctly** matched pair.
(1) Perinuclear space - 10 to 50 nm
(2) Width of chloroplast - 5 to 10 µm
(3) Length of a mitochondrion - 1.0 to 4.1 mm
(4) Length of mycoplasma - 0.3 nm

100. Due to a mitotic inhibitor, the cytoplasmic division did not take place after formation of daughter chromosomes in a cell. This will lead to;
(1) aneuploidy.
(2) formation of syncytium.
(3) immediate death of cell.
(4) restart of the cell cycle.

[Contd...

101. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Sweet potato and potato are example of convergent evolution in plants.
Reason R: Both represent stem modifications that have evolved to perform the function of food storage in completely different habitats.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

102. Which of the following compounds acts as a lectin?
(1) Vinblastin (2) Concanavalin A
(3) Anthocyanin (4) Ricin

103. Given below are two statements:
Statement I: At a chemical synapse, the membranes of the pre-synaptic and post-synaptic neurons are separated by the synaptic cleft.
Statement II: The transmission of an electrical impulse across an electrical synapse is always remarkably slower than across a chemical synapse due to the required diffusion of neurotransmitters.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

104. How many of the following genetic disorders have normal number of chromosomes?
A. Haemophilia
B. Sickle-cell anaemia
C. Down's syndrome
D. Phenylketonuria
E. Turner's syndrome
Choose the **correct** option.
(1) Two only (2) Three only
(3) Four only (4) Five

105. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The creation of genetically modified (GM) crops has successfully increased the efficiency of mineral usage by these tailored plants.
Reason R: Enhanced mineral usage fundamentally increases the early exhaustion of the fertility of the agricultural soil.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

106. What is the ploidy level and chromosomal count of a human secondary spermatocyte?
(1) Diploid, containing 46 chromosomes.
(2) Haploid, containing 23 chromosomes.
(3) Diploid, containing 46 chromatids.
(4) Diploid, containing 23 chromosomal pairs.

107. Which of the following statements about cockroach are **correct**?
A. Forewings (tegmina) arise from mesothorax and cover hind wings.
B. Hind wings arise from metathorax and are used in flight.
C. Female 7th–9th sterna form the brood pouch.
D. Males have short anal styles absent in females.\
E. Malpighian tubules are at foregut–midgut junction.
(1) A, B, C, and D only
(2) B, C, D, and E only
(3) A, C, and E only
(4) A, B, D, and E only

108. Radial symmetry is **not** found in which of the following taxa?
(1) Echinodermata
(2) Cnidaria
(3) Ctenophora
(4) Cyclostomata

[Contd...

109. Which statements about the human renal tubules are **correct**?

A. Descending limb of Henle: permeable to water, not electrolytes.

B. Ascending limb of Henle: impermeable to water, transports electrolytes.

C. PCT: lined by simple squamous epithelium for filtration.

D. DCT: secretes H^+ and K^+ to regulate blood pH.

E. Collecting duct: blocks urea passage to prevent toxicity.

(1) A, B, and D only

(2) A, C, and E only

(3) B, C, and D only

(4) A, B, D, and E only

110. Which of the following STI is classified as completely incurable even if detected early?

(1) Syphilis (2) Chlamydiasis

(3) Genital herpes (4) Trichomoniasis

111. Match **List-I** with **List-II** w.r.t frog.

List-I		List-II	
(A)	Testes	(I)	Yellowish and ovoid in shape
(B)	Ureters	(II)	Act as urinogenital ducts
(C)	Sinus venosus	(III)	Present on the ventral side of heart
(D)	Conus arteriosus	(IV)	Triangular structure

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-IV, B-II, C-III, D-I

(3) A-I, B-II, C-IV, D-III

(4) A-III, B-IV, C-I, D-II

112. Recombinant DNA is formed during meiosis at which stage?

(1) Metaphase I, when homologous chromosomes align at equator

(2) Anaphase I, when sister chromatids separate

(3) Prophase I, during crossing over between non-sister chromatids

(4) Telophase II, when four haploid cells are formed

113. Which pair is **incorrectly** matched?

(1) Neutrophils – phagocytosis of invading organisms

(2) Eosinophils – fight infections and mediate allergies

(3) Basophils – most abundant, secrete antibody-rich plasma

(4) Lymphocytes – mediate humoral and cell-mediated immunity

114. Which physiological conditions in tissues promote oxygen release from oxyhaemoglobin?

(1) High pO_2, low pCO_2, lesser H^+ concentration, and lower temperature.

(2) Low pO_2, high pCO_2, high H^+ concentration, and higher temperature.

(3) High pO_2, high pCO_2, neutral pH, and stable temperature.

(4) Low pO_2, low pCO_2, lesser H^+ concentration, and extreme low temperature.

115. What is the net gain of ATP molecules when one molecule of glucose is fermented to alcohol or lactic acid?

(1) 38

(2) 8

(3) 2

(4) 4

116. Identify the **incorrect** statement.

(1) Calvin mapped the pathway of carbon assimilation in photosynthesis after World War II.

(2) Penicillin was used extensively to treat American soldiers wounded in World War II.

(3) Bassham along with Calvin put radioactivity to beneficial use for studying sugar synthesis in green plants.

(4) The principles of photosynthesis as established by Ingenhousz are, at present, being used in studies on renewable resource for energy.

117. Which of the following are parasitic plants?

A. *Orobanche*

B. Venus flytrap

C. Pitcher plant

D. *Striga*

E. *Cuscuta*

Choose the **correct** option.

(1) A and B only (2) C, D and E

(3) A, B, C and D (4) A, D and E

[Contd...

118. Given below are two statements:
Statement I: Embryological support for evolution was also proposed by Ernst Heckel based upon the observation of certain features during embryonic stage common to all vertebrates that are absent in adult.
Statement II: Karl Ernst von Baer supported and proved Haeckel's proposal by demonstrating that embryos always pass through the exact adult stages of other lower animals.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

119. Match **List-I** with **List-II** w.r.t results of Mendelian dihybrid cross in pea in F_2 generation.

List-I (Trait)		**List-II (Proportion)**	
(A)	Round and green seeds	(I)	3/16
(B)	Non-parental traits	(II)	10/16
(C)	Parental traits	(III)	6/16
(D)	Wrinkled and green seeds	(IV)	1/16

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-II, B-IV, C-I, D-III
(4) A-III, B-I, C-IV, D-II

120. An example of artificial selection is;
(1) production of pink coloured flowers in *Antirrhinum* sp.
(2) no production of sugar by allele *i* of gene *I* in ABO blood group system of humans.
(3) pleiotropic effect of defect in gene that synthesises phenyl alanine hydroxylase.
(4) production of Sahiwal cows in Punjab.

121. Which of the following is a example of competitive inhibition?
(1) Inhibition of carbonic anhydrase by elevated intracellular zinc ions.
(2) Inhibition of the enzyme succinic dehydrogenase by malonate.
(3) Inhibition of RNA polymerase by excessive thermostable DNA primers.
(4) Inhibition of beta-galactosidase by chromogenic insertion.

122. Match **List-I** with **List-II** w.r.t type of placentation typically found in the given plant family.

List-I		**List-II**	
(A)	Compositae	(I)	Axile
(B)	Cruciferae	(II)	Parietal
(C)	Malvaceae	(III)	Basal
(D)	Leguminosae	(IV)	Marginal

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-IV, C-II, D-I
(2) A-III, B-II, C-I, D-IV
(3) A-I, B-II, C-III, D-IV
(4) A-IV, B-I, C-II, D-III

123. Given below are two statements:
Statement I: The tRNA is synthesised by transcription.
Statement II: In prokaryotes, tRNA cannot be synthesised.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

124. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: A person with blood group O is called
a Universal Donor.
Reason R: Group O blood lacks both A and B antigens on the surface of RBCs.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

125. Which of the following statements about animal phyla are **correct**?
A. Porifera: water enters via ostia, flows into spongocoel, exits through osculum.
B. Ctenophores: marine, diploblastic with comb plates for locomotion.
C. Platyhelminthes: triploblastic, pseudocoelomate with complete digestive tract.
D. Annelida: nervous system with paired ganglia and double ventral nerve cord.
E. Aschelminthes: dioecious, females often longer than males.
(1) A, B, C, and D only
(2) A, B, D, and E only
(3) B, C, D, and E only
(4) A, C, and E only

126. A population of finch birds with medium sized beaks colonizes an island with plants producing medium and large sized seeds. After a disease wipes out all the plants with medium sized seeds, what is most likely to happen to the finches?
(1) They will become extinct.
(2) Their beaks will not change at all.
(3) Their beaks become smaller.
(4) Their beaks will become larger.

127. An organism that can breed many times in a life cycle and produces small number of large-sized offsprings can be a/an;
(1) pelagic fish. (2) oyster.
(3) mammal. (4) pacific salmon fish.

128. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Robert May	(I)	Popularised the term biodiversity
(B)	Alexander Von Humboldt	(II)	Gave the concept of species-area relationship
(C)	Edward Wilson	(III)	Proposed the "rivet popper hypothesis"
(D)	Paul Ehrlich	(IV)	Gave an estimation of global biodiversity

Choose the *most appropriate* answer from the options given below.
(1) A-II, B-I, C-IV, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-I, B-II, C-IV, D-III
(4) A-II, B-III, C-I, D-IV

129. During an analysis of three cells, following observations were made.
(i) Very high number of nucleoli of large size
(ii) No compartmentalization of the cytoplasm
(iii) Cell wall present
Choose the option that is most suitable for the above observations.
(1) i- cell lacks ribosomes, ii- most likely a fungal cell, iii- cell wall contains galactans and mannans
(2) i- rRNA synthesis compromised, ii- glycogen synthesis at high rate, iii- lacks cell envelope
(3) i- cell is in interphase, ii- Ribosomal sub units are 40S and 60S, iii- cannot interact with neighbouring cells
(4) i- eukaryotic cell actively carrying out protein synthesis; ii- a prokaryotic cell, iii- can be a eukaryotic or a prokaryotic cell

130. Identify the **correct** statement about carotenoids.
(1) Present in P700 of light harvesting complex of chloroplast.
(2) Present in P680 of light harvesting complex of chloroplast.
(3) Stored in leucoplasts of a plant cell.
(4) Act as precursor for the stress hormone in plants.

131. Reabsorption of which of these molecules in kidney occurs via passive transport?
(1) Urea
(2) Glucose
(3) Sodium ions
(4) Amino acids

132. Which function is executed by the hormones secreted from the outermost layer of adrenal cortex?
(1) Regulating the balance of water and electrolytes in the body
(2) Stimulating massive gluconeogenesis and lipolysis
(3) Inducing the "fight or flight" emergency response by increasing alertness.
(4) Influencing the growth of axial, pubic, and facial hair during puberty

133. Identify the layer of connective tissue that encases a muscle fascicle?
(1) Sarcolemma
(2) Sarcoplasmic reticulum
(3) Fascia
(4) Myofibril

134. A doctor prescribes administration of cyclosporin A to a patient. Most likely, this patient;
(1) underwent myocardial infarction.
(2) has high blood cholesterol level.
(3) had an organ transplant.
(4) has deficiency of vitamin B_{12}.

135. The growth in a root elongating at a constant rate;
(1) initiates with a lag phase.
(2) has no parameter of measurement.
(3) is said to be determinate.
(4) occurs in an arithmetic manner.

[Contd...

136. Match **List-I** with **List-II**.

List-I		List-II	
(A)	EFB	(I)	Breaks open the bacterial cell wall to facilitate DNA extraction.
(B)	Traditional hybridisation	(II)	Inclusion and multiplication of undesirable genes
(C)	DNA Polymerase	(III)	Synthesizes continuous multiple copies of the gene of interest during PCR.
(D)	Lysozyme	(IV)	European Federation of Biotechnology

Choose the **correct** answer from the options given below:

(1) A-IV, B-II, C-III, D-I
(2) A-IV, B-I, C-II, D-III
(3) A-III, B-II, C-IV, D-I
(4) A-II, B-IV, C-I, D-III

137. Given below are two statements:
Statement I: GAA and GAG code for glutamic acid.
Statement II: In some nematodes, UUU does not code for phenylalanine.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

138. If the Hershey and Chase experiment is repeated with a bacteriophage having RNA labelled with radioactive phosphorus as genetic material, what would be the expected observation?

(1) DNA is the genetic material in bacteria.
(2) Bacterial DNA will not get incorporated with radioactive phosphorus.
(3) Radioactivity due to phosphorus will be detected in the supernatant.
(4) Radioactivity due to sulphur will be detected in the bacterial pellet.

139. In *Funaria*;

(1) gametophyte is dependent on sporophyte.
(2) meiosis takes place in the cells of secondary protonema.
(3) spores are produced in specialised structures called capsule.
(4) rhizoids are unbranched.

140. After a new action potential from the Sinoatrial Node, what occurs next in the cardiac cycle?

(1) AV valves open due to ventricular pressure
(2) Ventricles contract and semilunar valves close
(3) Atria contract, increasing ventricular filling
(4) Blood flows into left atrium through closed bicuspid valve

141. Cat and monkey belong to same ____ but different ____.

(1) class; orders
(2) kingdom; phyla
(3) species; genera
(4) family; classes

142. Why does over-secretion of Growth Hormone (GH) in adults not lead to a further increase in height?

(1) GH receptors are lost from the bone surface after puberty.
(2) The epiphyseal plates of long bones close after adolescence.
(3) Muscle growth inhibits the expansion of the skeletal framework.
(4) The liver stops producing growth hormones in middle age.

143. Given below are two statements:
Statement I: During non-cyclic photophosphorylation, plastoquinone gets reduced to plastoquinol after receiving electrons.
Statement II: The light reaction of photosynthesis occurs in the stroma of chloroplast.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

144. Which of the following statements is correct about a plant that produces only cleistogamous flowers?

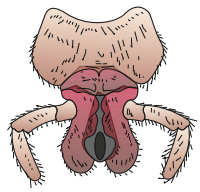
(1) It can produce only clones of itself.
(2) Availability of a pollinating agent increase chances of pollination in it.
(3) It can show the phenomenon of inbreeding depression.
(4) All its flowers are invariably unisexual.

[Contd...

145. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The inheritance of seed coat colour in an angiosperm does not depend on male parent's genes.
Reason (R): The alleles of a parental pair segregate from each other during gamete formation, such that a gamete receives only one of the two factors.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

146. Which of the following **accurately** differentiates the class Chondrichthyes from Osteichthyes?
(1) Terminal mouth, presence of a protective operculum, and air bladder.
(2) Ventrally located mouth, tough skin with minute placoid scales, and absence of an air bladder.
(3) Streamlined body, exclusively freshwater habitat, and a three-chambered heart.
(4) Cycloid scales, external fertilization, and highly specialized poison stings.

147. The mouth part of the cockroach shown in the given diagram acts as a/an:



(1) upper lip. (2) tongue.
(3) lower lip. (4) uvula.

148. What is the regulatory function of limbic system?
(1) Processing auditory inputs and maintaining severe postural balance.
(2) Controlling complex voluntary motor movements and fine muscle coordination.
(3) Regulating sexual behaviour, motivation, and the expression of emotional reactions like fear
(4) Directly maintaining cardiovascular reflexes and involuntary gastric secretions.

149. In an unknown plant, the tissue responsible for food transport was removed. The pathogen of which of the following diseases will not be able to spread within the plant beyond the point of tissue removal?
(1) Citrus canker (2) Curly top
(3) Tobacco mosaic (4) Bakanae

150. How is the respiratory exchange of gases carried out in adult frogs while they are in water?
(1) Through paired lungs
(2) Exclusively through internal gills.
(3) Through the skin by diffusion
(4) Through specialized spiracles and tracheoles.

151. A monocotyledonous stem lacks;
(1) vascular bundles.
(2) ground tissue.
(3) cork cambium.
(4) complex tissues.

152. In recombinant DNA technology, a reporter gene is often used along with a selectable marker to identify transformed host cells. Which of the following best explains the role of a reporter enzyme in this process?
(1) It kills non-transformed cells, allowing only transformed cells to survive.
(2) It provides antibiotic resistance to transformed cells.
(3) It produces a detectable signal (such as colour change or fluorescence) that confirms successful uptake and expression of foreign DNA.
(4) It enhances replication of plasmid DNA inside the host cell.

153. Choose the **incorrect** statement regarding different groups of Protozoans.
(1) Amoeboid protozoans capture their prey by putting out pseudopodia, and marine forms have silica shells.
(2) Flagellated protozoans are either free-living or parasitic, and the parasitic forms like *Trypanosoma* cause sleeping sickness.
(3) Ciliated protozoans are aquatic and have thousands of cilia for locomotion and steering food into the gullet.
(4) Sporozoans, like *Entamoeba*, have an infectious spore-like stage in their life cycle.

[Contd...

154. Identify the **incorrectly** matched pair.
(1) Ascomycetes - Sac-fungi, with septate and branched mycelium
(2) Basidiomycetes - Asexual spores are generally not found
(3) Deuteromycetes - Most members help in mineral cycling
(4) Phycomycetes - Mycelium is septate and coenocytic

155. Read the following statements.
A. A decomposer in an ecosystem has genetic material that may or may not be bound by nuclear membrane.
B. All decomposers are obligate aerobes.
C. Decomposers perform catabolism of detritus by breaking it down into smaller particles.
D. Digestive enzymes secreted by decomposers degrade detritus into complex inorganic substances.
E. The action of decomposers on detritus is independent of environmental conditions.
The **incorrect** statements are;
(1) A, B, C and D (2) B, C, D and E
(3) A, C, D and E (4) A, B, D and E

156. When a section of a population migrates to a completely new place multiple times this specific phenomenon is termed:
(1) Genetic drift
(2) Natural selection
(3) Gene flow
(4) Saltation

157. Which pair is **incorrectly** matched?
(1) Bone Marrow – primary organ producing all blood cells
(2) Thymus – lobed organ shrinking with age
(3) Spleen – secondary lymphoid organ, erythrocyte reservoir
(4) MALT – 20% of total lymphoid tissue

158. Match **List-I** with **List-II.**

	List-I		List-II
(A)	Oogonia	(I)	A large haploid cell formed by the unequal first meiotic division.
(B)	Primary Oocyte	(II)	Gamete mother cells formed during embryonic stage.
(C)	Secondary Oocyte	(III)	The fully mature tertiary follicle that ruptures during ovulation.
(D)	Graafian Follicle	(IV)	Cells temporarily arrested in the prophase-I stage of meiotic division.

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-I, C-IV, D-III
(4) A-III, B-IV, C-II, D-I

159. Match **List-I** with **List-II**.

	List-I		List-II
(A)	Multipolar neuron	(I)	Retina of the eye
(B)	Bipolar neuron	(II)	Usually in the embryonic stage
(C)	Unipolar neuron	(III)	Cerebral cortex
(D)	Myelinated nerve fibre	(IV)	Spinal and cranial nerves

Choose the **correct** answer from the options given below:
(1) A-III, B-II, C-I, D-IV
(2) A-III, B-I, C-II, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-IV, B-I, C-II, D-III

160. What key challenge did Eli Lilly overcome in 1983 to produce functional human insulin in *E. coli*?
(1) Synthesizing C peptide for folding
(2) Preventing pro-hormone degradation in bacteria
(3) Assembling A and B chains with correct disulfide bonds
(4) Making bacteria interpret the eukaryotic poly-A tail

161. After double fertilisation in angiosperms;
(1) the antipodals develop into seed coat.
(2) the central cell of embryo sac forms a nutritive tissue.
(3) the zygote develops into a triploid tissue.
(4) all the cells of embryo sac degenerate.

162. In glycogen, how are the terminal ends of the carbohydrate chain labeled?
(1) Right: non-reducing, Left: N-terminal
(2) Right: reducing, Left: non-reducing
(3) Left: C-terminal, Right: highly reduced
(4) Both ends are reducing

163. What exact valvular event is solely responsible for producing dub sound during cardiac cycle?
(1) The abrupt closure of both the tricuspid and bicuspid valves.
(2) The rapid opening of the atrio-ventricular valves due to atrial pressure.
(3) The closure of the semilunar valves as the ventricular pressure falls during diastole.
(4) The forceful opening of the semilunar valves during peak ventricular systole.

[Contd...

164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Free living nitrogen fixer	(I)	Algae
(B)	Symbiotic nitrogen fixer	(II)	*Glomus*
(C)	Form one of the components of mycorrhiza	(III)	*Azospirillum*
(D)	Form one of the components of lichen	(IV)	*Rhizobium*

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-IV, D-II
(2) A-IV, B-II, C-I, D-III
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-II, D-I

165. Which of the following plants is considered to be most advanced?
(1) *Polysiphonia* (2) *Polytrichum*
(3) *Wolffia* (4) *Salvinia*

166. Match **List-I** with **List-II.**

List-I		List-II	
(A)	Many bony fishes	(I)	Uricotelic (excrete uric acid paste)
(B)	Mammals	(II)	Ammonotelic (excrete ammonia mainly by diffusion)
(C)	Birds	(III)	Ureotelic (excrete urea processed by kidneys)
(D)	Glomerulus	(IV)	A tuft of capillaries formed by the afferent arteriole

Choose the **correct** answer from the options given below:
(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-I, D-III
(3) A-II, B-III, C-I, D-IV
(4) A-III, B-I, C-IV, D-II

167. Given below are two statements:
Statement I: Each human antibody molecule consists of four distinct peptide chains: two small heavy chains and two longer light chains.
Statement II: Antibodies circulate in the blood, providing protection known as the cellular immune response.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

168. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Living cells of a complex plant tissue	(I)	Bast fibres
(B)	Dead cells of a complex plant tissue	(II)	Hypodermis of dicot stem
(C)	Living cells of simple plant tissue	(III)	Hypodermis of monocot stem
(D)	Dead cells of simple plant tissue	(IV)	Xylem parenchyma

Choose the *most appropriate* answer from the options given below.
(1) A-III, B-I, C-II, D-IV
(2) A-IV, B-I, C-II, D-III
(3) A-I, B-II, C-III, D-IV
(4) A-II, B-IV, C-I, D-III

169. Given below are two statements:
Statement I: The thyroid gland is composed of follicles and stromal tissues, where thyroid follicle synthesises two peptide hormones.
Statement II: Insulin deficiency and/or insulin resistance result in a disease called diabetes insipidus.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

170. Troponin controls binding of actin to myosin motor, thereby regulating muscle contraction. Binding of which ions to troponin is crucial for this purpose?
(1) Na (2) K
(3) Mg (4) Ca

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Basal body of cilia	(I)	27 microtubules
(B)	Axoneme of flagella	(II)	80S
(C)	Eukaryotic ribosome	(III)	20 microtubules
(D)	Prokaryotic ribosome	(IV)	70S

Choose the *most appropriate* answer from the options given below.
(1) A-I, B-III, C-II, D-IV
(2) A-IV, B-II, C-III, D-I
(3) A-IV, B-I, C-II, D-III
(4) A-III, B-I, C-IV, D-II

[Contd...

172. A hypothetical plant species shows two flower colour phenotypes—red and white. Red is dominant over white. In a population, this flower colour locus was found to be in Hardy–Weinberg equilibrium. If 490 out of a total population size of 1000 plants were found to be white-flowered, how many of the red-flowered plants will be heterozygous?

(1) 700
(2) 90
(3) 580
(4) 420

173. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): The complex II of mitochondrial ETS does not transport protons across the membrane.

Reason (R): The complex II of mitochondrial ETS has a Fe-S centre.

In the light of the above statements, choose the *most appropriate* answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

174. Given below are two statements:

Statement I: The last step of decomposition of detritus is leaching.

Statement II: The first step of decomposition of detritus is humification.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

175. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Predators keep prey populations under control.

Reason (R): Predators help in maintaining species diversity in a community.

In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

176. Read the following statements.

A. Ecosystem varies greatly in size.
B. Nutrient cycles, food chains and webs are created as a result of the energy flows within the ecosystem.
C. Typically, an ecosystem has four structural features and two functional features.
D. Saprophytes are important components of an ecosystem.
E. The energy flow in an ecosystem does not follow the laws of thermodynamics.

In the light of above statements choose the **correct** answer from the options given below:

(1) B, C and E are correct; A and D are incorrect
(2) C, D and E are correct; A and B are incorrect
(3) A, B and D are correct; C and E are incorrect
(4) A, C and E are correct; B and D are incorrect

177. How many of the following are typically not seen in a isobilateral leaf?

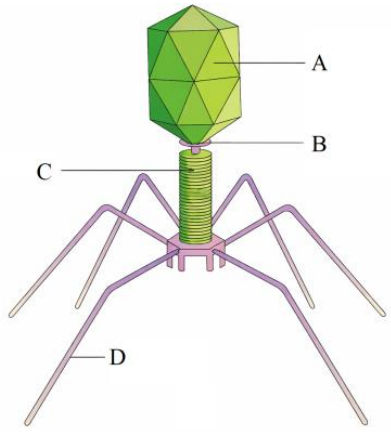
A. Adaxial epidermis
B. Bean shaped guard cells
C. Vascular bundles
D. Palisade parenchyma
E. Abaxial epidermis

Choose the **correct** answer.

(1) Two
(2) Three
(3) Four
(4) Five

[Contd...

178. Go through the organism shown below.



Identify the **correct** statement about it.

(1) It is considered to be non-cellular because it cannot produce its copies.

(2) Its DNA consists of 4.6×10^6 bp.

(3) The structure labelled as 'C' enters the host cell at the time of infection.

(4) It cannot be classified into any of the five kingdoms proposed by R. H. Whittaker.

179. Select the **correct** features of bean plants.

A. Flowers have five petals of variable size.

B. The seed retains endosperm at maturity.

C. The flower can be divided into two similar halves only in one particular vertical plane.

D. Stamens are arranged into two bundles.

E. Leaves are arranged alternatively with all veins of same size.

Choose the **correct** option.

(1) A, B and C (2) C, D and E

(3) A, D and E (4) A, C and D

180. Which of the following organisms exhibit Metagenesis, and what does it involve?

(1) *Aurelia*; polyps produce medusae sexually and medusae form polyps asexually.

(2) *Obelia*; polyps produce medusae asexually and medusae form polyps sexually.

(3) *Obelia*; medusae produce polyps asexually and polyps form medusae sexually.

(4) *Adamsia*; polyps produce medusae asexually and medusae form polyps sexually.

■■■

Kindly Share Your Feedback For this Paper

अपनी भावनाओं को समझें, तनाव और उलझनों से बाहर निकलें।

आप अकेले नहीं हैं – आपकी मदद के लिए PW प्रेरणा काउंसलर हमेशा तैयार हैं।

प्रेरणा हेल्पलाइन: +91 99905 00122

Discover Your Path to Emotional Wellness – Speak with a PW Prerna Counsellor Today!

PRERNA Helpline: +91 99905 00122



PW Web/App - https://smart.link/7wwosivoicgd4

Library- https://smart.link/sdfez8ejd80if



Believe in Yourself. Go Rock NEET 2026

PHYSICS WALLAH

Always Stands with You.